JAYESH MAGDUM

Python • Machine Learning • Data Analysis

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PROFESSIONAL SUMMARY

Final-year Information Technology engineering student with internship experience in Computer Vision, AI/ML, and robotics. Hands-on experience with Python, OpenCV, and YOLOv8 for industrial defect detection, object detection, and visual inspection. Also worked on AI-powered humanoid robotics, including speech processing and real-time person detection.

EXPERIENCE

Computer Vision Intern – Automators Industrial Projects (AIP) Oct 2026 – Present | Pune

- Visited the Brembo client site, tested industrial samples on an automated inspection machine, and worked with the client to validate defect detection results using 360° and 90° camera-based inspection.
- Contributed to an OAK-1 PoE camera-based hole detection system using YOLOv8, OpenCV, and DepthAI to detect hole centers and generate X–Y coordinates for robot-guided needle positioning.
- Supported a box-kit verification system that uses computer vision to detect components, identify missing parts, and determine OK/NOK assembly status.

Robotics Engineer Intern – MCAD Solutions Jul 2026 – Sep 2026 | Pune

- Integrated RAG-based responses, Speech-to-Text (STT), and Text-to-Speech (TTS) into Manas, an AI-powered humanoid robot, for conversational human-robot interaction.
- Implemented OpenCV-based person detection and face recognition for real-time interaction and personalized greetings.
- Contributed to robot integration and testing: synchronized neck and jaw movements during speech and basic pick-and-place operations.

AI / ML Intern – UptoSkills Feb 2026 – May 2026 | Remote

- Developed AI-powered features for a Smart CRM application, contributing to intelligent lead recommendations, chatbot integration, and AI-assisted business workflows.
- Performed AI service testing, API validation using Postman, and functional testing of chatbot responses, backend endpoints, and system integration.
- Conducted data preprocessing, exploratory data analysis (EDA), feature engineering, and machine learning model development using Python, Pandas, NumPy, and Scikit-learn.

PROJECTS

Manas – AI-Powered Humanoid Robot (MCAD) Python | OpenCV | RAG | STT/TTS

- Integrated RAG-based responses with Speech-to-Text (STT) and Text-to-Speech (TTS) for conversational human-robot interaction.
- Implemented real-time person detection and face recognition using OpenCV for personalized greetings; contributed to synchronized neck and jaw movements during speech.

Smart Waste Management and Reward System Python | OpenCV | Machine Learning

- Developed camera-based waste verification using OpenCV and QR codes, achieving 90%+ classification accuracy.
- Designed a points-based reward system for responsible disposal, reducing manual monitoring effort by ~60%.

Food Shelf-Life Prediction Python | Machine Learning | Data Analysis

- Trained a Random Forest model on a 5,000-record dataset to predict remaining food shelf life from food type, storage, temperature, humidity, and packaging.
- Developed an API to integrate the trained ML model and provide shelf-life predictions from input data.

TECHNICAL SKILLS

Languages: Python, SQL (MySQL), HTML/CSS

Computer Vision & AI: OpenCV, YOLOv8, Roboflow, Object Detection, Image Processing, Face Recognition, RAG, STT/TTS

ML / Data Science: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, TensorFlow, PyTorch

Concepts: EDA, Regression, Classification, Deep Learning, Predictive Modeling, Model Evaluation, Data Pipelines

Visualization & Tools: Power BI, Tableau, Excel, Streamlit, Git, GitHub, VS Code, Jupyter Notebook, DBMS

EDUCATION

B.E. – Information Technology — Marathwada Mitra Mandal College of Engineering, Pune 2023 – Present

Higher Secondary Certificate (XII) – Science — MIT Junior College 2023

Secondary School Certificate – CBSE — Podar International School 2021

ACHIEVEMENTS & CERTIFICATIONS

- SIH 2025 – Cleared Internal Round: Cleared the internal college round of Smart India Hackathon 2025 at MMCOE.
- Deloitte Australia and J.P. Morgan Job Simulations (Jan 2026): Built Tableau dashboards and Excel data models for forensic data analysis; performed Python-based financial data analysis and quantitative modeling.
- HackerRank Python Certification (Jan 2026): Demonstrated Python proficiency via standardized coding assessments.